AURA

Your home already senses you — privately.

Camera-free, sensor-free presence AI on a bare Arduino UNO Q

Competition	Arduino Physical AI Challenge India 2026 (Robu.in × Arduino × Qualcomm)
Category	Smart Homes & Consumer AI
Participant	Amalnath K P
Source code	`https://github.com/amalnathkp13-boop/aura` (open source, MIT)
Demo video	submitted alongside this report
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Bill of materials: one Arduino UNO Q. Nothing else.

1 · Executive summary

Smart homes want to know one thing above all — **is someone here, where, and are they okay?** — and today every mainstream answer compromises the home it serves: cameras watch your family, microphones listen to them, and every added sensor is more hardware, more cost, and usually another cloud subscription.

Aura answers the question with hardware the home already owns. The Arduino UNO Q's own WiFi radio hears the connected hotspot link and every access point around it; a human body moving through those radio paths bends them measurably. Aura converts a bare UNO Q — no camera, no microphone, no attached sensor of any kind — into a real-time presence, activity, and zone sensor with a live dashboard, an LED-matrix radar face, and Guardian modes that turn detection into consumer value: **Away** (instant Telegram intrusion alerts), **Home** (ambient awareness), and **Wellness** (inactivity alerts for elder care).

Everything runs on-device with **deterministic, explainable signal processing**: no training data, no neural black box, no cloud dependency — the system works with the internet down, and every decision traces to a threshold the user calibrated in their own room in fifteen minutes. The system is honest about its own limits by design: a calibration quality gate refuses a setup that cannot actually sense, and a drift detector raises a dashboard banner when the radio geometry changes rather than silently going blind.

0	1	~2 s	97
cameras, mics & attached sensors	item on the bill of materials	feature response to a doorway entry (replay-verified)	automated tests, all passing

100%

on-device — no cloud, no subscription

2 · The problem

Innovation & Originality

Occupancy is the primitive underneath most of the smart home: security arming, lighting and HVAC automation, and — increasingly important in India's multi-generational households — **wellness monitoring of elderly family members living alone**. The existing options each fail a different way:

Approach	What it costs the household
Cameras + vision AI	Constant surveillance of your own family; images of your home on someone's cloud; socially unusable in bedrooms and bathrooms — exactly where elderly falls happen.
PIR / mmWave / BLE sensors	A device per room, batteries, pairing, cost; PIR misses still people; another hub, often another subscription.
Wearables for elders	Only work when worn; the people who need them most forget or refuse them.

The result: the homes that most need presence awareness — for security while away and dignity-preserving care while home — are the least willing to install what it currently takes.

3 · The Aura idea

Innovation & Originality

The radio is the sensor. Radio waves at WiFi frequencies diffract around and reflect off the human body; a person crossing the path between the board and its hotspot — or between the board and any access point it can hear — measurably disturbs the received signal strength. Aura treats every audible radio path as an invisible tripwire and fuses them into one privacy-preserving presence sensor. It is *physically incapable* of taking a picture.

What is original here

- **A hard constraint as a product thesis:** the shipped system is a bare UNO Q — no peripheral of any kind. If Aura works, every UNO Q owner already owns the hardware. (The core RSSI feature extractor and rule classifier are adapted from an MIT-licensed open-source project — full attribution in `NOTICE.md` — everything else described below is Aura-original.)
- **Multi-link fusion:** instead of one radio link, Aura classifies every audible channel independently — connected-link RSSI at 32 samples/s plus each scanned AP — and fuses them with stability-weighted voting. A strict weighted majority is required for presence, so one glitching channel cannot raise a false alarm (measured: transmit-power steps on scan channels mimic entries ~2×/8 min; the fusion suppresses all of them).

- **Zone localization from signatures:** a two-minute per-spot calibration records how hard each channel is bent from that spot; at runtime the nearest log-scaled signature names the zone (`doorway`, `center`). Measured insight: two zones' signatures were 0.997-cosine parallel but 3× apart in magnitude — the magnitude *is* the zone information.
- **Self-aware calibration (fails loudly, not silently):** "Learn my room" derives per-channel thresholds with a quality gate that rejects a calibration that cannot see the user walk; at runtime a drift detector compares the live link level against its calibration-time anchor and raises a dashboard banner after 60 s of sustained >8 dB deviation while the room reads empty. This caught a real 19 dB hotspot relocation during our own validation day.
- **Deterministic, explainable consumer AI:** the dashboard shows the actual detector internals — per-channel votes, thresholds, live vote fractions, change-points — not a mascot animation. Users can see *why* Aura decided.
- **Dual-brain use of the UNO Q:** Linux side runs the sensing pipeline; the STM32 M33 real-time MCU renders a radar/aura animation on the 8×13 LED matrix, fed over the on-board Linux↔MCU bridge — the whole product in one device.

> **Honesty note.** Aura performs RF *sensing* with an imaging-*style* visualization. RSSI carries no position information and Aura never claims imaging, through-wall vision, or pose estimation. The radar view is deliberately abstract, and all accuracy claims in this report are our own measurements on this system.

4 · System architecture

Technical Documentation

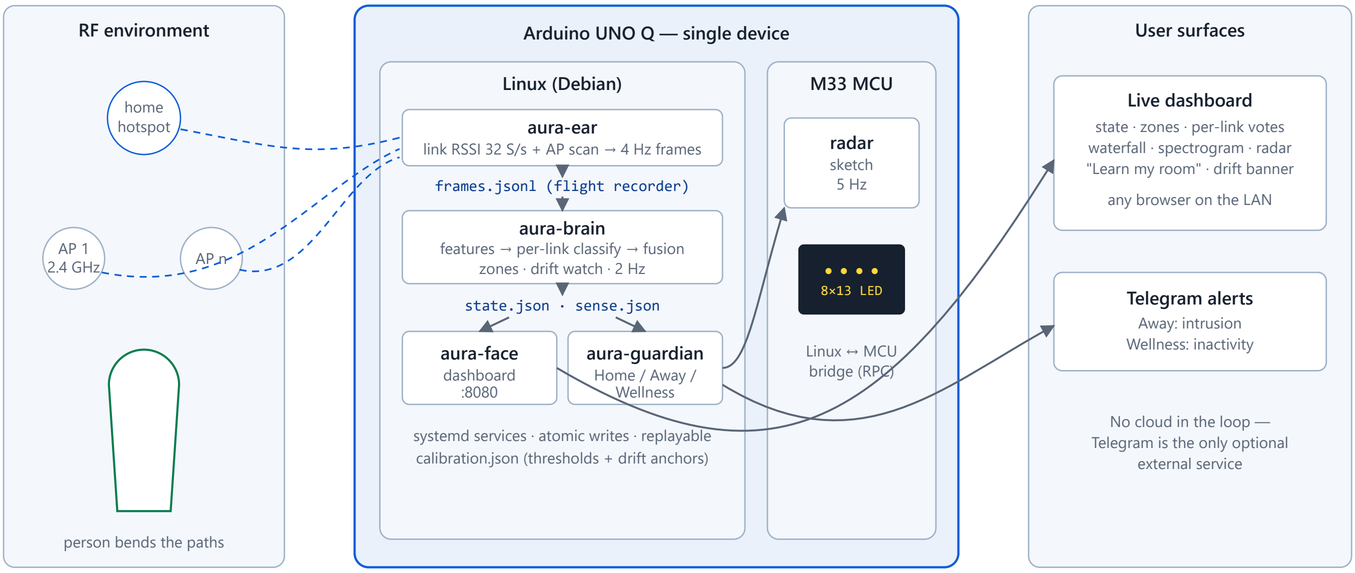

Figure 1 — Aura system diagram. A file-based pipeline of independent daemons on the UNO Q's Linux side, mirrored to the M33-driven LED matrix; every stage is restartable and replayable in isolation.

Design principles

- **Files as interfaces:** each stage communicates through append-only or atomically-replaced JSON files. `frames.jsonl` is a flight recorder: any live incident can be replayed offline through the exact production detector — this is how every number in §6 was verified.
- **Independent daemons:** `aura-ear` (radio capture), `aura-brain` (detection), `aura-face` (dashboard), `aura-guardian` (modes/alerts) are separate systemd services; a crash in one never takes down sensing.
- **Edge-only:** all computation on the QRB's Cortex-A53 cores; typical inference cost is a 15-s window of numpy signal processing every 500 ms.

5 · Detection pipeline

Technical Documentation

1. **Capture (4 Hz frames):** connected-link RSSI sampled at 32 S/s via station dump under a keep-alive ping (the sensing workhorse), plus periodic scans of every audible AP. MAC identifiers are salted-hashed at capture — raw addresses never leave the ear.
2. **Features (15-s sliding window, per channel):** RSSI variance; motion-band (0.5–2 Hz) and breathing-band (0.1–0.5 Hz) FFT power; CUSUM change-point count. A walking body produces oscillatory fading in the motion band; electronics produce steps and spikes — the features separate them.
3. **Per-channel classification** against thresholds calibrated for *this room*, with cross-receiver agreement between channels.
4. **Fusion (Aura-original):** stability-weighted voting across channels. Presence requires a strict weighted majority; motion likewise. Presence then holds 120 s past the last motion — a reading person is not an empty room.

5. **Zones:** per-spot calibrated signatures (median per-channel variance + band power); runtime nearest-match on log-scaled features with a distance ceiling and a runner-up margin, so an ambiguous reading yields *no* label rather than a wrong one.
6. **Calibration & self-monitoring:** "Learn my room" = 10 min empty + 5 min walking; per-channel thresholds derive from the empty distribution vs the walk distribution, and a **quality gate rejects** the calibration outright if walking is not separable from the empty floor (wrong placement). Each channel's calibration-time RSSI median is stored as a drift anchor; sustained deviation >8 dB for 60 s while the room reads empty raises the dashboard's **recalibrate banner**.

6 · Measured behaviour & validation

Functionality & Execution

All numbers below are measurements of this system in a real furnished room, using the production pipeline; replay-verified means the recorded frames were re-run offline through the exact deployed detector.

6.1 Measured engineering behaviour

Property	Measured value	Evidence
End-to-end plumbing latency (radio → dashboard)	≤ 1.5 s	4 Hz frames (0.17 s freshness measured), 2 Hz inference, 500 ms UI poll
Feature response to a doorway entry	~2 s	Replay: entry at t+0; channel votes *active* at t+2 s, variance 0.37 → 6.3 dB^2
Empty-room noise floor (link channel)	0.1–0.2 dB^2 variance	Live capture, quiet room, calibrated placement
Presence hold after last motion	120 s by design	Still-person test; replay-verified decay timing
False-alarm suppression by fusion	~2 single-channel glitch bursts per 8 min on an empty room — all suppressed	30-min empty capture replayed; glitches (AP transmit-power steps) never won the weighted majority
Calibration drift detection	Flags 60 s after a sustained >8 dB level shift	Caught a real 19 dB hotspot relocation on validation day; replay fires at exactly +60 s
Zone signature separability	Two real zones: cosine 0.997 in shape, 3× apart in magnitude	Live calibration captures; magnitude-aware matching implemented accordingly

6.2 Validation methodology (fully reproducible)

Aura ships its own honest benchmark harness rather than anecdotes. The protocol (`docs/validation-protocol.md`) scripts a live session — an empty-room phase, repeated doorway entries with stopwatch observation, a still-sitting phase, and continuous movement — with the ground-truth timeline declared at phase boundaries as absolute timestamps. The scorer (`training/validate.py`) then replays the recorded frames *chronologically* through the production detector (shuffled evaluation would break the deliberate 120-s presence hold) and reports:

- **Windowed presence accuracy** against the declared truth timeline;
- **Doorway-entry detection latency** — stopwatch-observed on the live dashboard, since windowed scoring has a 15-s floor by construction;

- **False-alarm windows per empty hour**;

each for both the fused detector and a naive single-threshold baseline, so the contribution of the fusion architecture is isolated rather than implied. The demo video shows the live system detecting in real time; anyone can reproduce the metrics table on their own room with the repository as shipped — we consider that reproducibility a stronger claim than a table measured once in one room.

6.3 Reliability engineering

- **97 automated tests** (pytest) across features, classification, fusion, calibration incl. the walk gate and drift detection, zones, services, and the dashboard API.
- **Replay harness:** any recorded session re-runs offline through the production detector — used throughout development for regression forensics.
- **Atomic state writes**, crash-isolated daemons, boot-persistent systemd units, one-command tar-based deploy.
- **Failure honesty:** the two failure modes we hit in the field — bad placement and post-calibration geometry change — are both now detected and reported to the user (walk gate; drift banner) instead of degrading silently.

7 · Smart-home & consumer value

Functionality & Execution

Mode	Consumer story
Away (security)	Arm when leaving; any detected presence sends an instant Telegram alert. A security system with nothing pointed at your family and nothing for an intruder to notice.
Wellness (elder care)	Inverted logic: *absence* of activity where there should be some raises the alert. Works in bedrooms and bathrooms — precisely where cameras are unacceptable and falls actually happen. No wearable to forget.
Home (ambient awareness)	Presence/zone state on the dashboard and LED matrix; a foundation signal for lighting/HVAC automation (occupancy-driven energy saving) via the same JSON state files.

Consumer-AI properties: fifteen-minute self-calibration with honest failure reporting; explainable decisions a user can inspect; no cloud account, no subscription, no data leaving the home (Telegram alerts are the single, optional, outbound integration); total added hardware cost of zero.

8 · Technical documentation

Technical Documentation

8.1 Bill of materials

#	Item	Qty	Role
1	Arduino UNO Q (Qualcomm QRB Linux SoC + STM32 Cortex-M33 MCU, on-board WiFi, 8×13 LED matrix)	1	The entire system: sensor, brain, display
—	*Already in the home:* WiFi hotspot/router; any browser on the LAN	—	Far end of the sensing link; user interface

Circuit schematic: none exists to draw — no external circuit is constructed, no pin is wired, no peripheral is attached. The board is used exactly as shipped; Figure 1 (system diagram) documents the complete signal path in lieu of a schematic. This is the point of the project, not an omission. **The Arduino UNO Q is not merely the primary board — it is the only device in the system.**

8.2 Repository map

`aura/ear`	Radio capture daemon (link RSSI + AP scan → 4 Hz frames; salted-hash anonymization)
`aura/brain`	Feature extraction, per-link classification, multi-link fusion, zones, drift watch, calibration
`aura/face`	Flask dashboard + RF sensing console (live detector internals)
`aura/guardian`	Home/Away/Wellness modes, Telegram alerting
`board-app/aura-matrix`	M33 LED-matrix radar app (Linux↔MCU bridge)
`training/validate.py`	Chronological scorer: frames + truth timeline → metrics table
`tests/ (97 tests)`	Unit + service tests for every stage
`docs/`	Validation protocol, data log, design specs
`deploy/`	One-command deploy + systemd units

8.3 Licensing, originality & compliance

MIT-licensed, fully open source. Aura is an original, previously unpublished project, built solo for this challenge. In the spirit of good open-source practice we disclose proactively: the low-level RSSI feature extractor and rule classifier (two files) are adapted from an MIT-licensed upstream project, with the pinned upstream commit, file-level provenance, local modifications, and full license text recorded in `NOTICE.md`. Every other component — the capture pipeline, multi-link fusion, zone localization, drift detection, calibration quality gates, dashboard and RF console, guardian modes, validation tooling, and the M33 LED-matrix app — is original to Aura and written for this submission.

9 · Honest limitations & roadmap

- **Sensing coverage follows radio geometry:** sensitivity concentrates around the board↔hotspot path. Aura turns this from a silent failure into a guided setup: the walk gate rejects bad placement and the dashboard's drift banner demands recalibration when geometry changes.
- **Windowed confirmation:** features respond in ~2 s, but confirmed presence uses a 15-s analysis window — Aura is a presence system, not a millisecond tripwire.
- **RSSI, not CSI:** the UNO Q's radio exposes signal strength, not channel state information; Aura therefore claims presence/activity/zones — never imaging or pose.
- **Roadmap:** Home Assistant / Matter bridge from the existing JSON state; multi-board room mesh with cooperative fusion; breathing-band stillness detection for sleep-safe wellness; CSI-capable radio support where hardware allows.

10 · Submission artifacts

Source code (open source, MIT)	`https://github.com/amalnathkp13-boop/aura`
Demo video (3–5 min)	submitted with this report
System diagram	Figure 1 (also provided as a standalone image)
UNO Q purchase proof	attached separately, as required